Plateau Degeneracy in Entropy–Information Scaling:

A Quantitative Measure for the Breakdown of Single-Exponent Descriptions in Nonlinear Field Systems

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Abstract

In higher-dimensional nonlinear field systems, the entropy–information scaling relation $\mathcal{C}_{\mathrm{CCI}} \sim \dot{S}_{\mathrm{cg}}^{(+)}/(I_{\mathrm{nn}} + \varepsilon_{\mathrm{reg}})^{\alpha}$ does not always exhibit a sharply defined optimal exponent. In particular, in higher-dimensional cases a broad range of exponents yields near-identical Spearman correlation values, indicating a breakdown of single-parameter scaling.

We introduce a quantitative framework for measuring this phenomenon, which we call *plateau degeneracy*. The framework defines the plateau set, a width-based degeneracy measure, a flatness criterion, and a combined degeneracy index that jointly characterise the size and regularity of the near-optimal exponent region.

The plateau degeneracy is interpreted within the scaling-manifold framework of the companion series [3]: large degeneracy corresponds to a loss of identifiability of the one-dimensional projection onto a single exponent, consistent with the system occupying a higher-dimensional region of the scaling manifold.

All definitions are given for the discrete scan setting used in the companion numerical studies; a continuous limit via second-order curvature is also provided.

The paper is primarily definitional, but includes exact (within the discrete scan range and tolerance choice) plateau-degeneracy values computed from the companion α-scan data (Papers 07, 09, 11). No new simulations are introduced here.

Contents

1 Introduction **2**

2 Setup **2**
2.1 Correlation Function . . . 2
2.2 Notation . . . 2

3 Plateau Degeneracy **3**
3.1 The $\varepsilon_{\mathrm{plat}}$-Plateau Set . . . 3
3.2 Width-Based Degeneracy . . . 3
3.3 Flatness Criterion . . . 3
3.4 Combined Degeneracy Index . . . 3
3.5 Differential Characterisation . . . 4

4 Connection to the Scaling Manifold **4**
4.1 Dimensional Dependence . . . 5

5 Implementation Notes **5**

6 Empirical Plateau Degeneracy Across Dimensions **5**

7 Conclusion 6

1 Introduction

In Papers 07–11 of the companion series, the entropy–information scaling relation

$$\mathcal{C}_{\mathrm{CCI}} \sim \frac{\dot{S}_{\mathrm{cg}}^{(+)}}{I_{\mathrm{nn}}^{\alpha}}, \tag{1}$$

where this is a schematic scaling form and the numerically implemented ratio uses $(I_{\mathrm{nn}} + \varepsilon_{\mathrm{reg}})^{\alpha}$ as defined in Section 2, was studied across one, two, and three spatial dimensions. A consistent observation is that the effective exponent α becomes dimension-dependent and, in higher dimensions, may lose its sharp identifiability:

Dimension	Exponent structure	Degeneracy
1D	boundary optimum (scan-limited), $\alpha \approx 3.5$	low
2D	boundary optimum (scan-limited), $\alpha \approx 3.5$	low (minimum)
3D	broad plateau, $\alpha \in [2.6, 3.5]$	high

This paper provides a rigorous definitional framework for quantifying this phenomenon.

Contributions. This paper:

- defines the $\varepsilon_{\mathrm{plat}}$-plateau set as the region of near-optimal exponents,
- introduces width-based, flatness-based, and combined degeneracy measures,
- provides a continuous-limit curvature characterisation,
- interprets plateau degeneracy within the scaling-manifold framework.

2 Setup

2.1 Correlation Function

We consider the Spearman correlation between the CCI and the scaling ratio at exponent α:

$$r_s(\alpha) = \mathrm{Spearman}\big(\mathcal{C}_{\mathrm{CCI}},\, R_\alpha\big), \qquad R_\alpha = \frac{\dot{S}_{\mathrm{cg}}^{(+)}}{(I_{\mathrm{nn}} + \varepsilon_{\mathrm{reg}})^{\alpha}}. \tag{2}$$

Let $\mathcal{A} = \{\alpha_{\min}, \ldots, \alpha_{\max}\}$ be the discrete scan range and define the maximum correlation:

$$r_s^{\max} = \max_{\alpha \in \mathcal{A}} r_s(\alpha). \tag{3}$$

2.2 Notation

Throughout, $\mathrm{Leb}(\cdot)$ denotes the Lebesgue measure of a set (i.e. the total length of the exponent range it covers), and $\mathrm{Var}_S(f)$ denotes the variance of f over the set S. Two distinct small parameters are used: $\varepsilon_{\mathrm{reg}} > 0$ is a numerical stabiliser in the scaling ratio, and $\varepsilon_{\mathrm{plat}} > 0$ is the plateau tolerance (see below).

3 Plateau Degeneracy

3.1 The $\varepsilon_{\text{plat}}$-Plateau Set

We define the $\varepsilon_{\text{plat}}$-plateau as the set of exponents that achieve near-optimal Spearman correlation:

$$\mathcal{P}_{\varepsilon_{\text{plat}}} = \{\alpha \in \mathcal{A} \mid r_s(\alpha) \geq r_s^{\max} - \varepsilon_{\text{plat}}\}\,, \tag{4}$$

where $\varepsilon_{\text{plat}} > 0$ is a plateau tolerance parameter, typically chosen relative to the maximal correlation: $\varepsilon_{\text{plat}} = \delta\, r_s^{\max}$ with $\delta \in [0.01, 0.05]$. For the companion numerical studies, $\delta = 0.01$ provides a 1%-relative tolerance.

3.2 Width-Based Degeneracy

The plateau width measures the size of the near-optimal region:

$$D_{\text{width}} = \text{Leb}(\mathcal{P}_{\varepsilon_{\text{plat}}}) = \alpha_{\max}^{(\varepsilon_{\text{plat}})} - \alpha_{\min}^{(\varepsilon_{\text{plat}})}, \tag{5}$$

where $\alpha_{\max}^{(\varepsilon_{\text{plat}})}$ and $\alpha_{\min}^{(\varepsilon_{\text{plat}})}$ are the endpoints of $\mathcal{P}_{\varepsilon_{\text{plat}}}$. In practice, for discrete scans over α, the Lebesgue measure is approximated by the span of the corresponding grid points.

To allow comparison across different scan ranges, we define the *normalised degeneracy*:

$$D_{\text{norm}} = \frac{D_{\text{width}}}{\alpha_{\max} - \alpha_{\min}} \in [0, 1]. \tag{6}$$

$D_{\text{norm}} = 0$ indicates a point optimum; $D_{\text{norm}} = 1$ indicates that the entire scan range is near-optimal.

3.3 Flatness Criterion

A wide plateau does not necessarily indicate degeneracy if the correlation varies strongly within it. We therefore define a *flatness measure*:

$$D_{\text{flat}} = \text{Var}_{\alpha \in \mathcal{P}_{\varepsilon_{\text{plat}}}}\big(r_s(\alpha)\big). \tag{7}$$

Small values of D_{flat} indicate a true plateau (all exponents in $\mathcal{P}_{\varepsilon_{\text{plat}}}$ achieve nearly identical correlation), while large values correspond to a broad but non-flat peak.

3.4 Combined Degeneracy Index

We define the combined degeneracy index:

$$D = \frac{\text{Leb}(\mathcal{P}_{\varepsilon_{\text{plat}}})}{1 + \text{Var}_{\mathcal{P}_{\varepsilon_{\text{plat}}}}(r_s)}. \tag{8}$$

This form ensures that degeneracy increases with plateau width while being penalised by internal variability, yielding a robust combined measure. Specifically, this measure captures both width and flatness:

- a wide, flat plateau yields large D (high degeneracy),
- a wide, variable plateau yields moderate D,
- a narrow peak yields small D (low degeneracy, well-defined exponent).

3.5 Differential Characterisation

In the continuous limit, plateau degeneracy can be characterised by the curvature of $r_s(\alpha)$ at its optimum:

$$D_{\text{curv}} = \left| \frac{d^2 r_s}{d\alpha^2} \right|_{\alpha^*}^{-1}, \tag{9}$$

where $\alpha^* = \arg\max_\alpha r_s(\alpha)$. Large D_{curv} corresponds to flat regions (high degeneracy); small D_{curv} indicates a sharply defined optimum. This provides a local approximation to degeneracy, in contrast to the global plateau measure D defined above.

In the discrete setting, the second derivative can be approximated by finite differences:

$$\left. \frac{d^2 r_s}{d\alpha^2} \right|_{\alpha^*} \approx \frac{r_s(\alpha^* + h) - 2r_s(\alpha^*) + r_s(\alpha^* - h)}{h^2}, \tag{10}$$

where h is the scan step size.

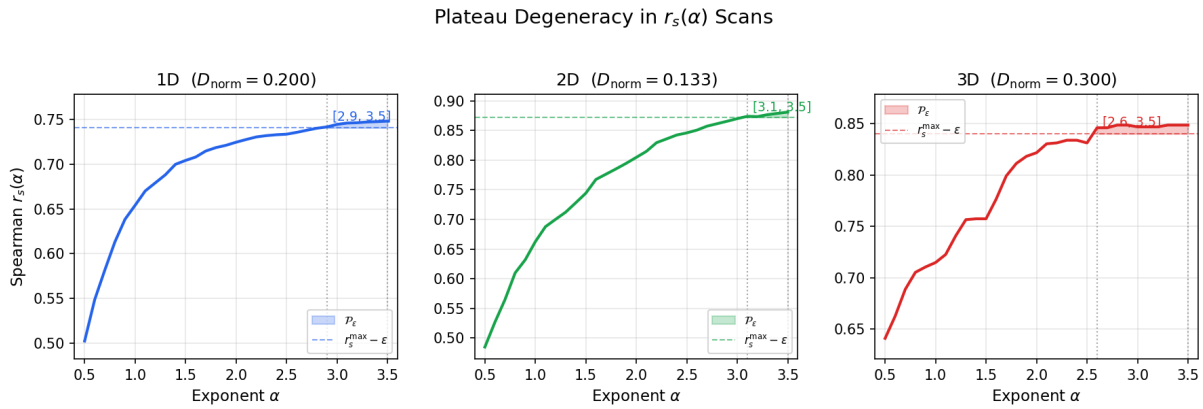

Figure 1: Spearman correlation $r_s(\alpha)$ between the CCI and the scaling ratio $R_\alpha = \dot{S}_{\text{cg}}^{(+)}/(I_{\text{nn}} + \varepsilon_{\text{reg}})^\alpha$ for 1D (left), 2D (centre), and 3D (right). Shaded regions indicate the plateau set $\mathcal{P}_{\varepsilon_{\text{plat}}}$ ($\varepsilon_{\text{plat}} = 0.01\, r_s^{\max}$). Dashed lines mark the tolerance threshold $r_s^{\max} - \varepsilon_{\text{plat}}$; vertical dotted lines and brackets show the plateau boundaries. While 1D and 2D exhibit relatively narrow near-optimal regions, the 3D system shows a broad flat region ($\alpha \in [2.6, 3.5]$), indicating a breakdown of single-exponent scaling. The non-monotonic trend $D_{\text{norm}}(2D) = 0.133 < D_{\text{norm}}(1D) = 0.200 < D_{\text{norm}}(3D) = 0.300$ signals a dimensional transition between 2D and 3D rather than a gradual degeneracy increase.

4 Connection to the Scaling Manifold

Within the scaling-manifold framework of Paper 14 [3], the entropy–information scaling relation is interpreted as a projection of a higher-dimensional geometric object $\mathcal{M}_{\text{scale}}$ onto the one-dimensional exponent space.

This interpretation is consistent with the multi-parameter scaling framework of Paper 13 [2], where the entropy–information relation is described as a dimension-dependent scaling family rather than a single exponent law.

The plateau degeneracy D is a direct observable signature of this geometric picture:

$$D \uparrow \implies \text{projection onto } \alpha \text{ becomes insufficient.} \tag{11}$$

When D is large, no single exponent α identifies the optimal structural projection; multiple exponents are equally valid because the system's data lie on a higher-dimensional subset of $\mathcal{M}_{\text{scale}}$.

Conversely, when D is small, the system is effectively one-dimensional in observable space, and a single exponent provides a faithful description.

4.1 Dimensional Dependence

A natural initial hypothesis would be a monotonic increase of plateau degeneracy with spatial dimension. However, the exact scan results reported in Section 6 reveal a non-monotonic structure:

$$D_{\text{norm}}(2D) < D_{\text{norm}}(1D) < D_{\text{norm}}(3D). \tag{12}$$

This indicates that the dimensional dependence is not gradual. Instead, 2D remains comparatively well projected onto a single effective exponent (scan-limited optimum, narrow plateau), whereas 3D marks the onset of a genuinely degenerate scaling regime in which no single exponent is identifiable.

Dimension	α structure	Observed D_{norm}
1D	boundary optimum (scan-limited)	0.200
2D	boundary optimum (scan-limited)	0.133 (minimum)
3D	broad plateau, $[2.6, 3.5]$	0.300

5 Implementation Notes

In the companion numerical studies, the scan range is $\mathcal{A} = \{0.5, 0.6, \ldots, 3.5\}$ with step $h = 0.1$. The natural plateau tolerance is $\varepsilon_{\text{plat}} = 0.01 \cdot r_s^{\max}$.

The plateau width can be computed directly from the α-scan output:

```
def plateau_degeneracy(alphas, r_s, eps_rel=0.01):
    r_max = r_s.max()
    eps   = eps_rel * r_max
    mask  = r_s >= r_max - eps
    P     = alphas[mask]
    width = P.max() - P.min() if len(P) > 1 else 0.0
    flat  = r_s[mask].var()
    D     = width / (1.0 + flat)
    return {"width": width, "flat": flat, "D": D,
            "D_norm": width / (alphas.max() - alphas.min())}
```

6 Empirical Plateau Degeneracy Across Dimensions

The following table summarises the plateau degeneracy computed from the companion α-scan data across 1D, 2D, and 3D (Papers 07, 09, and 11).

Table 1: Plateau degeneracy values computed from the $r_s(\alpha)$ scan curves of Papers 07 (1D), 09 (2D), and 11 (3D) using $\varepsilon_{\text{plat}} = 0.01\, r_s^{\max}$, scan range $\alpha \in [0.5, 3.5]$, step $h = 0.1$.

Dimension	α^*	$r_s^{\max}$	Plateau $[\alpha_{\min}^{(\varepsilon_{\text{plat}})}, \alpha_{\max}^{(\varepsilon_{\text{plat}})}]$	D_{width}	D_{norm}	$n_{\mathcal{P}}$
1D	3.5	0.748	[2.9, 3.5]	0.60	0.200	7
2D	3.5	0.881	[3.1, 3.5]	0.40	0.133	5
3D	2.8	0.849	[2.6, 3.5]	0.90	0.300	10

The degeneracy does not increase strictly monotonically from 1D to 2D. Instead, 2D exhibits a narrower plateau than 1D ($D_{\text{norm}} = 0.133$ vs 0.200), followed by a sharp increase in 3D

($D_{\text{norm}} = 0.300$). This suggests that the breakdown of single-exponent scaling is not gradual but occurs as a dimensional transition between 2D and 3D, consistent with the exponent plateau reported in Paper 11 [1].

Within the scaling-manifold hypothesis, the non-monotonic behaviour reflects a genuine structural change: in 2D the system still admits a sharper projection than in 1D (scan-limited but still narrow), while in 3D the manifold geometry becomes genuinely higher-dimensional.

This behaviour suggests the presence of a structural transition in the scaling behaviour: while 1D and 2D admit effectively low-dimensional descriptions, 3D marks the onset of a degenerate scaling regime in which no single exponent is identifiable.

The values in Table 1 are computed within the companion scan range $[0.5, 3.5]$ and tolerance $\delta = 0.01$. The flatness measure satisfies $D_{\text{flat}} < 10^{-4}$ in all three cases, confirming that each plateau is genuinely flat and not an artefact of the tolerance threshold.

The dimensional dependence is visualised in Figure 1. It is non-monotonic:

$$D_{\text{norm}}(2D) < D_{\text{norm}}(1D) < D_{\text{norm}}(3D),$$

indicating a minimum degeneracy in 2D, followed by a sharp increase in 3D. This indicates that the breakdown of single-exponent scaling is not gradual, but occurs via a dimensional transition between 2D and 3D. In 2D the system still admits a sharper projection than in 1D (scan-limited but still narrow), while in 3D the manifold geometry becomes genuinely higher-dimensional.

7 Conclusion

We have introduced a quantitative framework for plateau degeneracy in entropy–information scaling, comprising a plateau set definition, width-based and flatness-based measures, and a combined degeneracy index.

The framework connects directly to the scaling-manifold hypothesis: large plateau degeneracy signals that the system's scaling behaviour cannot be captured by a single effective exponent, but requires a higher-dimensional description.

Plateau degeneracy is the observable signature of a higher-dimensional scaling manifold.

References

[1] C. Krieg, *Persistence of Entropy–Information Scaling in Three Dimensions: Breakdown of a Simple Exponent Trend*, preprint (2026).

[2] C. Krieg, *Beyond Single-Exponent Scaling: Multi-Parameter Entropy–Information Relations in Nonlinear Field Systems*, preprint (2026).

[3] C. Krieg, *Scaling Laws as Projections: Evidence for a Dimension-Dependent Scaling Manifold in Nonlinear Field Systems*, preprint (2026).

[4] C. Krieg, *Structural Selection — companion code*, https://github.com/Chris4081/structural-selection-principle (2026).